LAST PAPER / RESEARCH EVIDENCE REVIEW

Detailed Experiment Explanation

From the five-branch structural-credit audit to E1B, E2, E3, E4, and patch-level E3 v2/v3

Purpose. A researcher-oriented learning document: precise experimental questions, statistical mechanisms, configurations, quantitative results, failure analysis, limitations, and decisions. Written to support independent reasoning rather than imply that the full proposed system has already been validated.

Evidence cut-off: 12 October 2026. Primary numerical evidence: supplied five-branch/E4/patch E3 audit JSONs. E1B, E2, and original image-level E3 results: the 7 October research handoff and repository protocol documents. Code versions are cited in the reference inventory.

Status key: OBSERVED = completed empirical result; IMPLEMENTED / NOT RUN = code exists but no experiment result was supplied; PROPOSED = future experiment, not performed.

Question	Current answer	Strength
Does correct selective credit help source routing?	Yes, under verified-address, useful-feedback controls.	OBSERVED / controlled
Do rare mask corrections persist?	Small positive reuse under random sparse masks.	OBSERVED / handoff
Do 32-D address sidecars beat raw DINO?	No tested sidecar passed.	OBSERVED / handoff
Does NIG help sparse normal density?	Large calibration benefit vs simple plug-in Gaussian; little AUROC difference.	OBSERVED / image-pooled
Does E4 already improve the detector?	No supported end-to-end benefit in v1.	OBSERVED / negative
Does multimodal local patch memory win?	No in first six-category CV fold.	OBSERVED / preliminary

Scope note: The earlier beta-VAE and Stage-1 factor-discovery audits are treated as background because this document begins with the five-branch controlled comparison. This is not a complete history of all preceding research.

1 Research question and non-negotiable experimental boundaries

The intended system observes a product, detects suspicious image regions through a frozen visual backbone, assigns responsibility to a small number of downstream memory addresses, receives sparse human feedback, and modifies only the relevant persistent external states. The backbone is not fine-tuned in these experiments. This is a narrower target than building a full autonomous active-inference agent.

> Frozen image → DINO patch features → anomaly Sensor → address / responsibility → verified feedback → permitted memory write → future prediction

1.1 Frozen infrastructure

• Dataset: Real-IAD, 30 product categories; five outer folds of six target categories each. Outer fold 0 targets: audiojack, bottle_cap, button_battery, end_cap, eraser, fire_hood. The other 24 categories are source categories, split into four non-overlapping groups of six for source cross-validation.

• Backbone: DINOv2 with registers, ViT-L/14, 1,024-dimensional fused patch features; selected blocks 4–18, mean fusion. Image processing: resize 448, center crop 392; 28 × 28 = 784 patch positions. No online change to DINO weights.

• Standard frozen detector: four-shot normal reference bank with cosine nearest-neighbour anomaly score. In E2 and image-level E3, the Sensor selects top-8 anomalous patches and computes a score-softmax weighted evidence vector (temperature 20).

• Source-CV categories are legitimate mechanism-development data; the six outer-fold target categories must not be used to select models, prior settings or query policies. Final outer-fold target rotation has not been performed.

1.2 Terminology that must not be conflated

Term	Operational meaning
Oracle evidence	Patch information selected using offline ground-truth mask: an upper-bound diagnostic, not a deployable sensor.
Sensor evidence	Patch information selected by the frozen anomaly detector without GT masks.
Address / factor	A named or indexed destination for a memory write; semantic/causal disentanglement has NOT been shown.
Normal state	Predictive distribution over verified-normal evidence; may be image-pooled or patch-level depending on E3 version.
Router / source bank	Matches observed anomaly evidence or responsibility signatures to Core-4 codes AK, HS, QS, ZW.
Query event	A selected inspection for human correction; oracle / fixed / random / known-normal schedules are scientifically different.
Sentinel	Held-out evaluation observations that never write to memory.
Image AUROC vs source macro-F1	Different endpoints: binary normal/defect ranking vs four-way defect-source routing.

1.3 How to read the evidence

A result can show that a mechanism works when given informative labels without demonstrating that the deployed sensor can obtain those labels. Likewise, an accurate source router can improve its own F1 while a completely disconnected binary detector remains unchanged. For each experiment, ask: What was allowed to change? What supervision was revealed? Which metric can change by construction?

2 Experimental chronology and dependencies

Sequence	What was isolated	Outcome
Five-branch structural credit	Correct-address vs no write, matched broadcast, full broadcast, shuffled.	Strong routing gain when true address is supplied.
E1B: rare masks	Reusable spatial positive / negative correction under sparse random masks.	Small reproducible improvements; controller deferred.
E2: address sidecars	32-D structured and linear address representations vs raw DINO.	All tested sidecars failed routing gate.
E3 (original)	Image-pooled sparse normal-state density; deterministic vs NIG / variants.	NIG calibration advantage; no large ranking gain.
E4 v1	Asymmetric normal and defect-source feedback paths under random queries.	No successful detector gate; small source-routing change.
Patch E3 v2	Patch scoring with 8 fixed feature-coordinate groups.	Both learn somewhat; no clear NIG advantage.
Patch E3 v3	Global 1-mode vs local 8-mode appearance memories.	Local variants trail global mean AUROC.
Patch E4 v2 / E3-D	Patch-feedback variants and diagnostic replay.	E4 v2 IMPLEMENTED/NOT RUN; E3-D PROPOSED.

These are different experimental models. In particular, the phrase “E3 NIG is selected” belongs to original image-pooled E3; it must not be silently generalized to the later patch-local multimodal setting.

3 Five-branch experiment — the decisive structural-credit control

3.1 Purpose and hypothesis

The first hard causal question was not “Can the system discover the correct factor by itself?” It was: if correct factor-address information is supplied, does writing to that address preserve useful evidence better than distributing the same feedback across all factors? A correct-address branch that wins against both mass-matched and full broadcast, while a wrong-address branch collapses, demonstrates that write location matters in this controlled regime.

3.2 Fixed protocol

• Outer fold 0; 24 source categories rotated into four source-CV held-out groups. Ten repeats per source-CV group = 40 correlated source-side runs per representation.

• Core-4 label set: AK, HS, QS, ZW. One initial labelled factor support per source. Fixed shared diagonal background statistics; factor memory stores effective count n_g and weighted sum for its shrunk mean.

• For both Oracle/raw patch and Sensor/raw top-8 (score-softmax) representations, the same stream, support state, 32 random query positions, and read-only sentinels are shared across branches. The controlled feedback is deliberately useful for Core-4 source learning.

• Checkpoints at 0, 4, 8, 16, 32 feedback events. Primary endpoint: future held-out Core-4 routing macro-F1. Two scorers: diag_shrunk and Bayesian predictive; their final argmax outcomes were very similar.

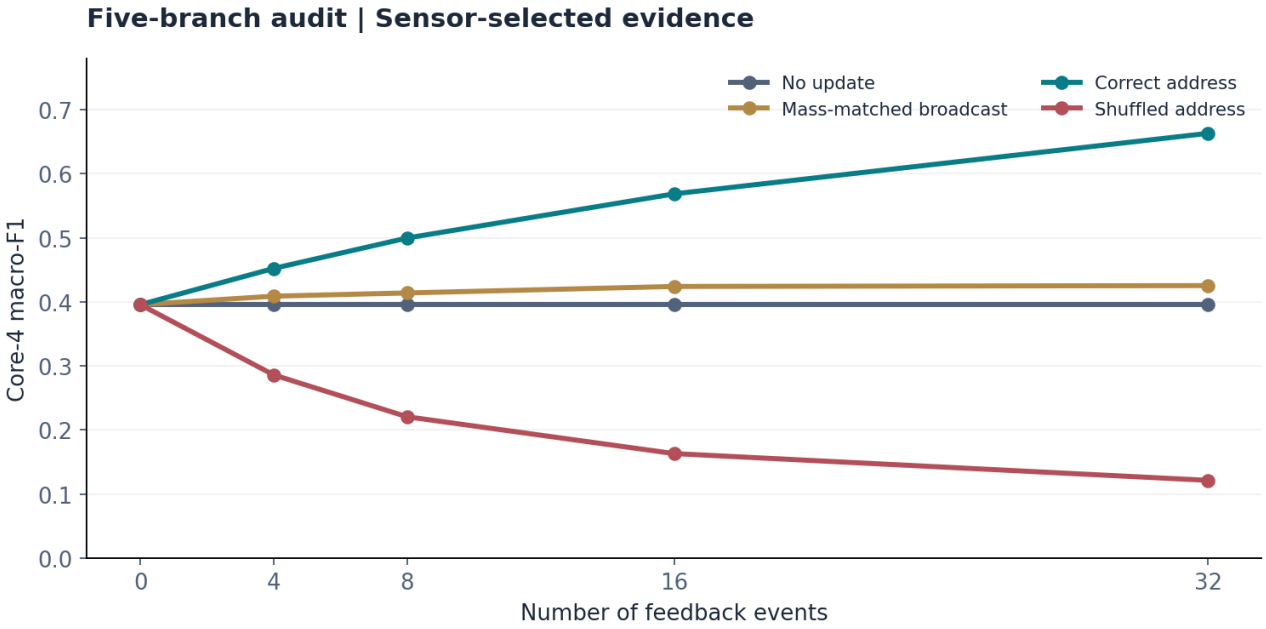
3.3 The five write rules

For feedback item x at verified source g^*, let credit $\alpha(g)$ determine each factor's update mass.

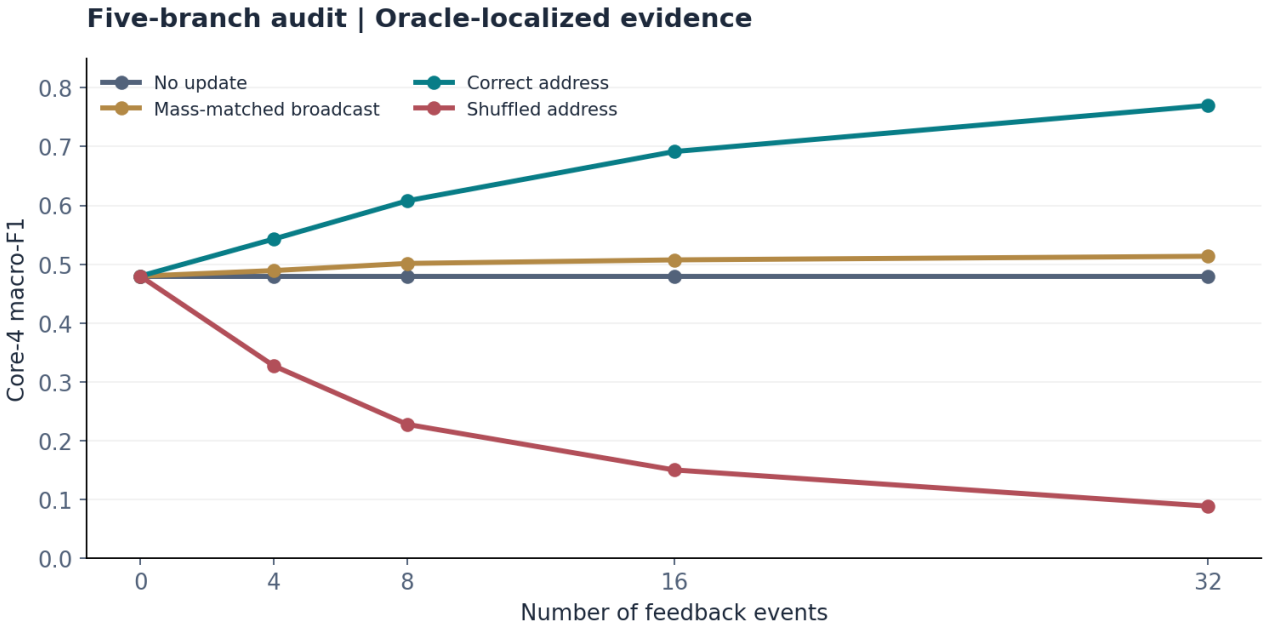
Branch	Credit assignment	Total mass
No update	$\alpha(g) = 0$ for every g.	0
Correct address	$\alpha(g^*) = 1$; $\alpha(g \neq g^*) = 0$.	1
Mass-matched global	$\alpha(g) = 1/4$ for each of four factors.	1
Full broadcast	$\alpha(g) = 1$ for each factor.	4
Shuffled address	$\alpha(\pi(g^*)) = 1$ for fixed derangement π.	1

Comparing against mass-matched broadcast isolates address from total write mass. Comparing against full broadcast tests the original "write all memories" idea with four times greater total mass. Shuffling destroys address correctness while keeping mass and feedback indices matched.

3.4 Results: source routing macro-F1



Primary reported branch: diag_shrunk and Sensor-selected raw-DINO top-8 evidence. A clear divergence is visible by Q16; Q32 correct-address = 0.6631, global broadcast = 0.4254, shuffled = 0.1215.



Oracle-localized raw patches are stronger, but the same ordering holds: correct-address = 0.7702 at Q32 vs mass-matched global = 0.5135.

Q	No update	Correct	Global mass=1	Full mass=4	Shuffled
0	0.3957	0.3957	0.3957	0.3957	0.3957
4	0.3957	0.4522	0.4089	0.4091	0.2860
8	0.3957	0.4997	0.4140	0.4077	0.2207
16	0.3957	0.5686	0.4241	0.4244	0.1632
32	0.3957	0.6631	0.4254	0.4259	0.1215

At Q32 with Sensor features, correct minus no-update = +0.2673 macro-F1, correct minus mass-matched global = +0.2376, correct minus full broadcast = +0.2371; shuffled falls 0.2743 below no-update. Source-side state-locality integrity passed. Across repetitions these are correlated runs, not 40 independent product generalizations.

3.5 What was proved; what was not

Supported: Controlled correct structural credit strongly outperforms indiscriminate/wrong allocation; correct persistent factor statistics can be updated locally, with no accidental update to other factor states. Not supported: correct factor labels being inferred from sparse image-only feedback, discovery of semantic factors, robust online query selection, factor-wise normal-state update, or downstream binary detector improvement. The true address here is revealed to the evaluator/learner. This is an important oracle-assisted mechanism proof, not the complete ABMG system.

4 E1B — sparse mask feedback can improve future patch selection

4.1 Question and mechanism

After demonstrating structural address credit under controlled labels, E1B asked whether a rarely requested full pixel-level defect mask can leave reusable state in a simple category-local correction memory. The GT mask is exceptionally rich feedback and only sporadically revealed in the experimental mask schedule. The frozen Sensor and DINO are not retrained.

The correction bank contains a small set of positive defect prototypes and negative/background prototypes. Subsequent patch scores receive local similarity-based correction. This is a spatial-evidence correction mechanism, not a Bayesian NIG normal-state update.

4.2 Conditions and measured gains

The handoff reports random mask schedules at 0.25%, 0.5% and 1.0% with suppress-only and bidirectional positive/negative correction. A primitive 8-positive/8-negative prototype bank saturates quickly; the 0.5%-to-1% increment is small.

Condition	Hit@8	Precision@8	Recall@8	Routing F1
Frozen Sensor	.9079	.4714	.5297	.6007
Suppress 0.5%	.9137	.4776	.5360	.6085
Bidirectional 0.5%	.9202	.4862	.5457	.6148
Bidirectional 1.0%	.9217	.4883	.5476	.6151

Relative to frozen Sensor, bidirectional 0.5% yields about +1.23 percentage points Hit@8, +1.47 pp Precision@8, +1.60 pp Recall@8, and +1.41 pp routing F1. Five schedule seeds retained small positive gains; source-normal diagnostic FPR decreased by roughly 1.82 ± 0.21 pp. The “51+ items persistence” bucket does not measure isolated one-mask causal lifetime, because another same-category mask may intervene.

4.3 Interpretation and deferred controller question

Conclusion: rare mask information has modest, reproducible persistent value in this primitive bank. The originally proposed high-spatial-entropy mask-query controller (B2) and its read-only comparator (B3) were intentionally deferred. Selecting masks adaptively would confound query quality with the learned correction state; a read-only branch with no current-item effect is otherwise structurally identical to no-write in a prequential future-only evaluation.

Interpretation boundary: These E1B numbers come from the 7 October handoff rather than a directly supplied raw E1B result JSON. The FPR is an E1B-specific diagnostic, not the official Stage-0 operating FPR.

5 E2 — do structured sidecars produce better addresses?

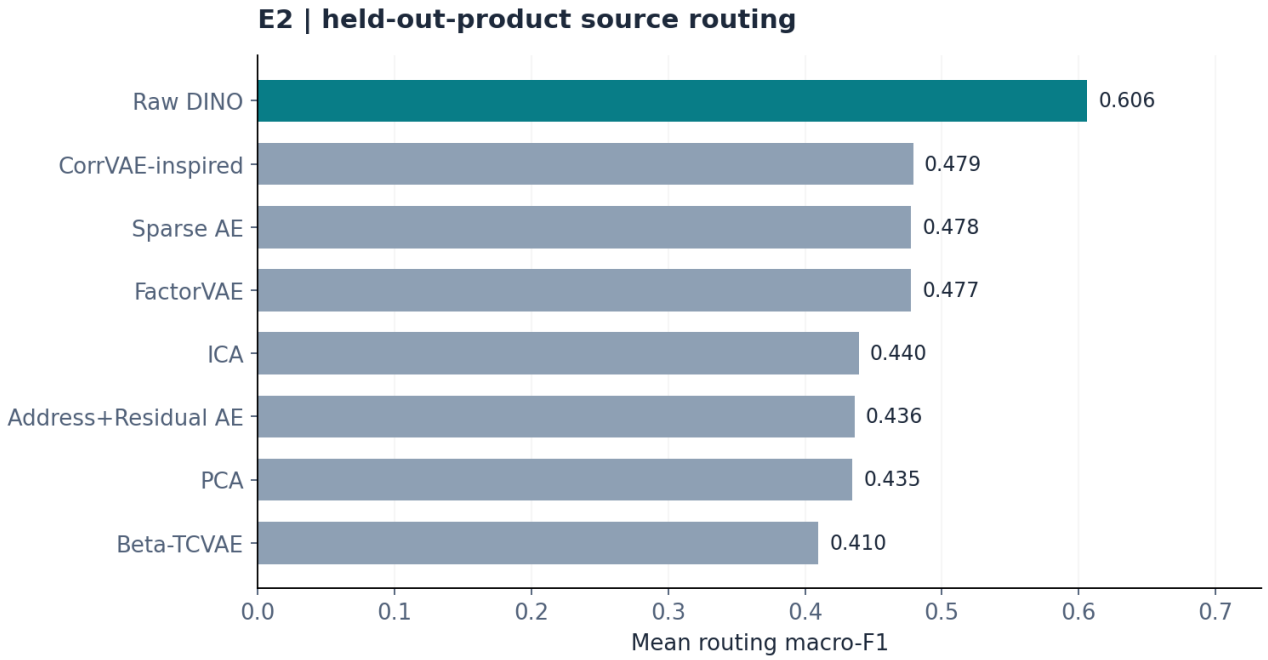
5.1 Why a downstream sidecar was tested

The intended architecture separates raw visual evidence h (used for anomaly scoring) from an optional compact address code z = f(h) (used only for routing / deciding where feedback belongs). Disentanglement, sparsity, and grouped correlation constraints might make memory more addressable, but compression can also destroy the very details that identify a defect source across products.

Every E2 candidate saw the same DINO patches, raw-DINO top-8 selection and spatial weights, and the same shared-diagonal shrunk routing memory. No model was permitted to replace the deployed detector or use target defects to train the sidecar.

5.2 Tested representations and protocol

Seven 32-dimensional candidates: PCA, ICA, Beta-TCVAE, FactorVAE, CorrVAE-inspired (8 groups × 4 coordinates, with cross-group covariance penalty/group sparsity), address+residual autoencoder, and sparse autoencoder. The raw 1024-dimensional DINO representation served as baseline. Source-CV: 18 products to fit on normal TRAIN features; six held-out products to assess Core-4 routing. A candidate needed +0.02 mean Sensor macro-F1, positive gains in ≥3/4 CV groups, and ≤0.02 Oracle F1 loss.



No candidate passed the E2 primary gate. The 32-dimensional representation budget was common across sidecars but not matched to the 1024-dimensional raw baseline.

Raw DINO scored Sensor macro-F1 0.6062 and Oracle macro-F1 0.7598. Best learned candidate CorrVAE-inspired reached only 0.4791 and 0.5988, respectively. No sidecar surpassed raw DINO in any of the four CV groups.

5.3 Why good patch separation did not imply good addresses

Several sidecars yielded true-factor patch AUROC approximately 0.91–0.93 while four-way source-routing F1 was poor. These are not contradictory: once a factor identity is already supplied, local defect/background separation can be strong, yet the representation may erase consistent differences between the four source identities. CorrVAE group participation was ~7.1/8 groups, not the desired sparse set. Sparse AE measured zero near-zero coordinates under its configured threshold. Some PCA information loss was expected: at 32 dimensions it retained only about 63% of normal-patch variance.

Decision: Retain raw DINO for current address routing. Do not claim factorized representations cannot work: E2 did not match raw/sidecar dimensionality or prove causal factor semantics. A graded PCA dimensionality control is a future diagnostic rather than proof against all sidecars.

6 Original E3 — image-level Bayesian normal-state density

6.1 What normality was being modeled

Each image was represented by a single top-8 score-weighted, L2-normalized 1024-D image descriptor. That vector was not a collection of 784 patch-level normal observations. A four-shot detector continued to identify suspicious patches; the 1/2/4-shot E3 condition specified separate support for the Bayesian normal-state model. This difference is crucial.

E3-A compared a deterministic plug-in diagonal Gaussian, a weak-prior diagonal Normal-Inverse-Gamma (NIG) with Student-t prediction, and a source-derived hierarchical NIG. E3-B injected 10% deliberately mislabeled defect vectors into verified-normal updates, adding a robust generalized-Bayes NIG weighting rule. All shared the same normal state support, update stream and held-out probes.

6.2 NIG model: variables, derivation and update

For a single feature coordinate j, let x denote a verified-normal scalar, μ its unknown normal mean and σ^2 its unknown variance. The observation model is $x \mid \mu,\sigma^2 \sim \text{Normal}(\mu,\sigma^2)$. The prior is Normal-Inverse-Gamma, parameterized by μ_0 (prior mean), $\kappa_0>0$ (prior mean precision measured as pseudo-count), $\alpha_0>1$ (inverse-gamma shape) and $\beta_0>0$ (inverse-gamma scale).

$$\mu \mid \sigma^2 \sim \text{Normal}(\mu_0, \sigma^2 / \kappa_0); \sigma^2 \sim \text{InvGamma}(\alpha_0, \beta_0).$$

Given n verified-normal observations $x_1,\ldots,x_n$, define the sample mean $\bar{x} = (\Sigma x_i)/n$ and scatter $S = \Sigma(x_i - \bar{x})^2$. Multiplying the Gaussian likelihood by the NIG prior and completing the square yields:

$$\kappa_n = \kappa_0 + n; \mu_n = (\kappa_0 \mu_0 + n \bar{x}) / \kappa_n$$

$$\alpha_n = \alpha_0 + n/2; \beta_n = \beta_0 + S/2 + [\kappa_0 n / (2 \kappa_n)](\bar{x} - \mu_0)^2$$

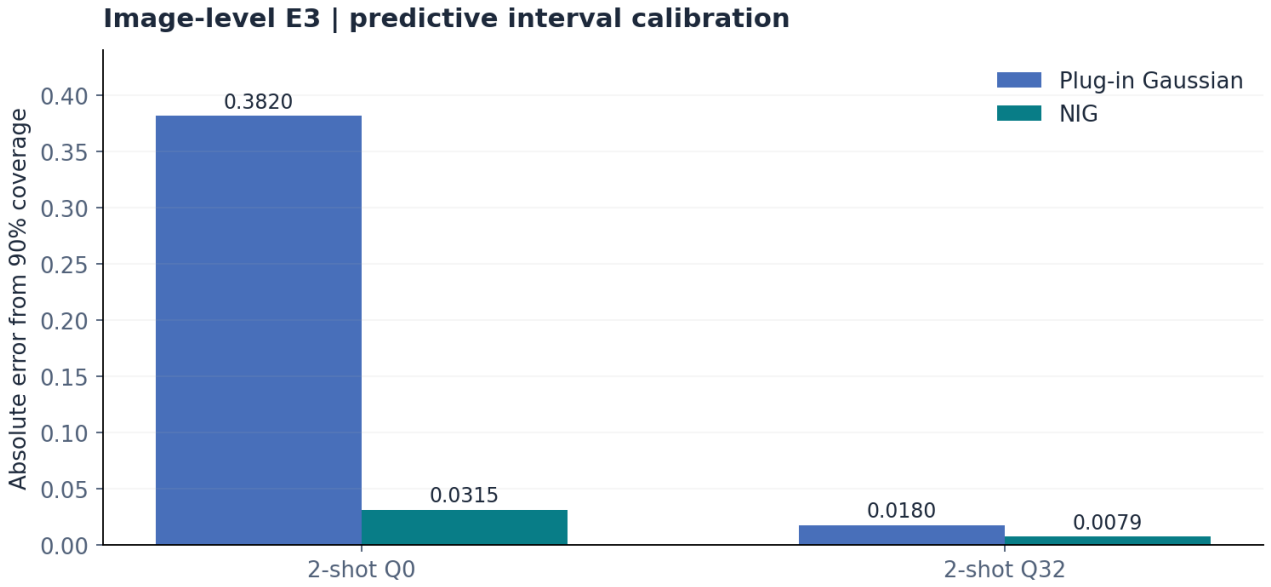
The next normal observation is integrated over both uncertain parameters, producing a Student-t predictive density with degrees of freedom $\nu = 2\alpha_n$, location μ_n and scale squared $s^2 = \beta_n(\kappa_n+1)/(\alpha_n \kappa_n)$. The anomaly surprise is the negative predictive log density, usually averaged across coordinates:

$$a(x) = -(1/D)\,\Sigma[j=1..D]\ \log \text{StudentT}(x_j \mid \nu_j, \mu_{n,j}, s_j).$$

Here D=1024 in the original E3. The mean of coordinate-wise log densities is a diagonal approximation, not a full correlated joint model. When observations carry fractional weights, n becomes effective mass, $\bar{x}$ a weighted mean and S a weighted scatter; the same algebra applies to power-likelihood updates. Patch-wise correlation makes such fractional updates an approximation in later E3 versions.

6.3 Why predictive calibration matters independently of AUROC

Binary AUROC measures the relative ranking of defect and normal images; it is invariant to strictly increasing score transforms. 90% predictive-interval calibration asks whether held-out normal coordinates fall inside nominal 90% intervals about 90% of the time. “Calibration error” is the absolute difference between empirical coverage and 0.9. Good AUROC does not guarantee honest uncertainty and vice versa.



Original image-level E3: two-shot NIG predictive intervals are much better calibrated than a simple plug-in Gaussian. This plot is not patch-level E3 v2/v3.

At 2-shot, Q0: deterministic AUROC 0.71494, NIG 0.71724; deterministic 90% interval coverage 0.51798 (error .38202), NIG coverage 0.86934 (error .03154). At Q32: AUROC 0.76311 vs 0.76203, calibration error .01800 vs .00789. NIG passed the predeclared non-inferiority margin $\Delta\text{AUROC} \geq -.01$ and calibration benefit gate; it did not demonstrate large ranking gains or clearly reduced support-seed AUROC variability.

6.4 Alternative NIG configurations

Hierarchical NIG: source-derived coordinate-wise κ_0 about 4.70–4.92 (median) versus .01 for weak NIG, giving stronger transfer but less ability to adapt to an unseen product. It lost about .0165 AUROC at 2-shot Q0 and .0182 at 2-shot Q4 vs ordinary NIG. Lower posterior drift was partly unwelcome inertia.

Robust NIG: a bounded residual-dependent influence weight was intended to suppress bad writes. However, 10% corrupted updates still received roughly .933–.936 influence (clean ~.976), and damage

was only ~.0007–.0010 AUROC smaller than ordinary NIG. The tested robust weighting was not an adequate write-authority mechanism. A mislabeled defect written into normality is treated as normal by NIG; Bayesian uncertainty does not automatically verify labels.

6.5 Valid decision and scope limitation

Decision in original E3: choose ordinary NIG/Student-t as a sparse image-level normal model versus the tested simple plug-in Gaussian. Do not extrapolate superiority to stronger shrunk deterministic models, patch-level multimodal observations, or factor-addressed normal-state updates. The next patch experiments explicitly changed the statistical problem.

7 E4 v1 — asymmetric verified feedback under random queries

7.1 Why E4 was harder than the five-branch proof

E4 began to test the interface between normal feedback, defect-source feedback, and future prediction under a random-query stream. Unlike five-branch, queries were not selected to guarantee informative Core-4 corrections. Source-CV groups (four) × support seeds (three) yielded 12 correlated run outcomes, not 12 independent categories. Per run: 32 random query positions, checkpoints 0/4/8/16/32, normal support 4-shot, read-only held-out sentinels. The patch cache held only the top 64 frozen-Sensor candidates.

7.2 E4A v1: verified normal, N0 versus N1

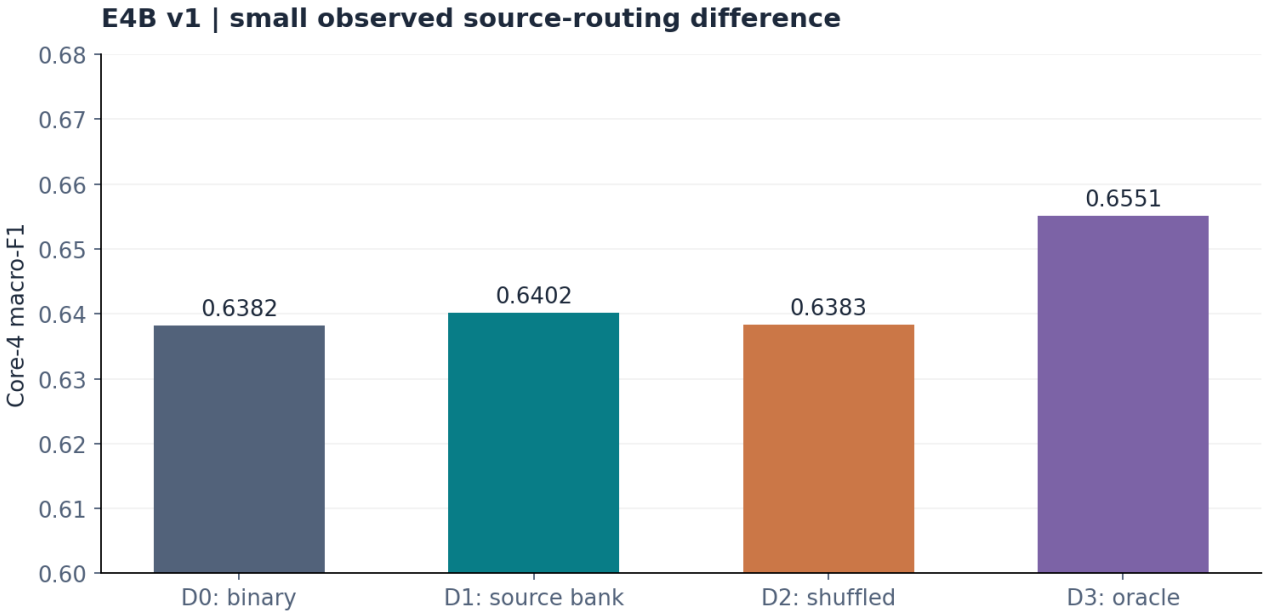
N0 updates the category normal-state NIG with verified normals but has no hard-normal correction bank. N1 performs the identical NIG update; a verified false positive additionally writes the frozen top-8 patch descriptors to a small category-local negative correction bank. Future candidate patch scores may be adjusted and top-8 reranked within only the cached top 64. This is not factor-specific Bayesian NIG memory and cannot recover patches ranked below the top-64 cutoff.

E4A Q32 metric	N0	N1	N1 − N0
Normal FPR	0.04427	0.04601	+0.00174
Defect recall	0.17238	0.17337	+0.00099
Image AUROC	0.73771	0.73853	+0.00082
Routing macro-F1	0.63824	0.63762	-0.00062

Normal FPR got worse by ~0.17 percentage points (positive N1−N0), while AUROC changed only +.00082. N1 thus did not meet the proposed E4A gate of at least 1 percentage-point FPR improvement without meaningful recall/AUROC loss. Its gains were not robust across the four CV groups; feedback opportunities were sparse and uneven.

7.3 E4B v1: source-enriched defect feedback

The normal-state model was fixed to N0. The router used Core-4 shrunk Gaussian state and initial uncalibrated softmax responsibilities r. For verified false-negative Core-4 defects, a source-centroid correction could shift fractional credit toward the previously observed defect-source signature; true positives could enrich source summaries without writing factor memory. Four policies: D0 binary-only responsibility r; D1 0.5r + 0.5q_source; D2 similarly mixed with shuffled source; D3 diagnostic one-hot true source credit.



Q32 held-out Core-4 routing macro-F1. D1–D0 = +0.00196; D1–D2 = +0.00187; both far below the proposed +0.02 gate. D3 oracle = 0.6551.

Crucial disconnection: in E4B v1, factor-source feedback changes the router but does not modify the binary normal/defect score. Therefore binary detector AUROC, normal FPR and defect recall are identical across D0–D3 by construction. These identical numbers cannot be used to argue that source-enriched feedback helps or fails at improving binary detection. The E4B result specifically evaluates a small source-routing effect.

7.4 Why E4 v1 is not a decisive refutation of ABMG

E4A had a low starting FPR (~.044) and its hard-normal bank used few useful updates; E4B used random queries and a routing-only write path, not the intended coupled spatial-factor Bayesian normal memory. Thus E4 v1 failed its implemented gates, but did not directly test the full spatially selective Bayesian write hypothesis. The reverse error is also unacceptable: success in five-branch cannot be used to claim E4 already works.

8 Why patch-level E3 was reopened

The original E3 was a statistics comparison on image-pooled evidence. The target architecture, however, should preserve patch identity and allow human feedback to update the relevant subset of normal appearances. The E4 results highlighted the mismatch: an image-level NIG mean has no inherent mechanism to identify which local patch appearance was corrected. Thus patch-level normal-state modeling is a new, non-equivalent E3 problem, not an automatic refinement of the prior positive NIG result.

8.1 What the full-patch cache contains

A new source-only frozen-feature extraction saved 3,696 images: 768 additional train normals, 1,536 test normals, 1,392 defect images. Every image has 784 ranked 1024-D patch descriptors, with detector fingerprint ec0dfc39c0a5ab7e. No GT defect masks are used for the normal density scorer. All six fold-0 target products remain sealed.

The extractor stored patches sorted by descending frozen Sensor anomaly score. Consequently it did not preserve original (row,column) patch indices. Full patch coverage is available, but 2D spatial locality cannot be recovered from the saved array without additional extraction metadata. This limits claims about spatially resolved Bayesian memory.

9 Patch E3 v2 — group-wise patch scoring

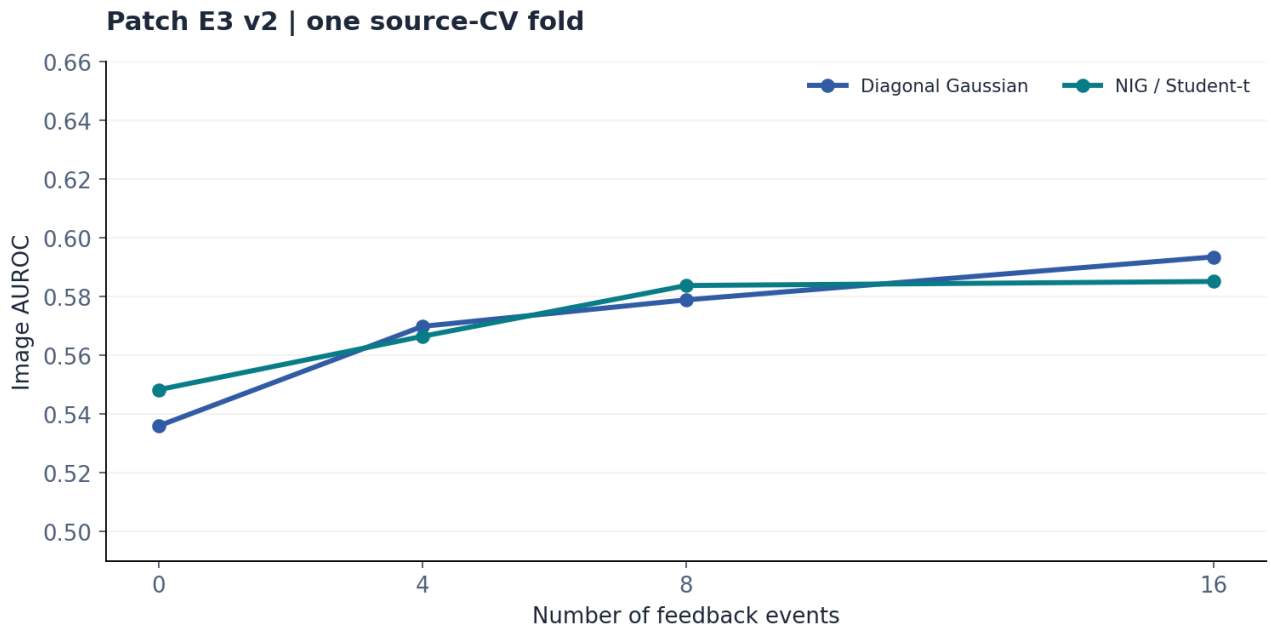
9.1 Mechanism

V2 partitions the 1024-D raw DINO vector into eight fixed 128-dimensional coordinate groups. Within each group it fits one diagonal Gaussian or one diagonal NIG normal-state model, using initial normal support and fractional batch statistics across rank-sampled patches. A verified normal image contributes an effective mass of one per group in this E3 experiment. For an evaluation image, it computes per-patch negative log density within each group, averages the top-eight patch surprises within each group, standardizes them by development priors and takes the maximum across groups.

Architectural issue: despite per-patch scoring, each group still contains only one unimodal density pooling together many unrelated patch appearances. “Eight groups” are feature coordinate slices, not learned appearance modes and not semantic factors. Moreover, the scorer uses the first 64 score-ranked patches from the 784-cache in actual E3 v2 evaluation, even though the extraction contained all patches.

9.2 Configuration and results

Outer fold 0, source-CV group 0 (mint, mounts, pcb, phone_battery, plastic_nut, plastic_plug), seed 0. Two normal-state support images, 16 controlled offline verified-normal queries chosen by cold-start NIG surprise, same query identities for both models, held-out normals/defects as sentinels. “Verified normal” is known offline here; this is not active human query selection.



Mean over six source products. Q16 Gaussian 0.5935 and NIG 0.5851; both gain from Q0, but the NIG advantage is not evident.

Q	Gaussian AUROC	NIG AUROC
0	0.5360	0.5483
4	0.5699	0.5665
8	0.5789	0.5838
16	0.5935	0.5852

Both models improve mean image AUROC with the selected normal stream (Gaussian +.0575; NIG +.0368), but Q16 NIG trails Gaussian by .0084. This does not contradict the original image-pooled E3 calibration success: the features, likelihood assumptions, parameterization and measured metric are different. V2 does not establish that patch-wise NIG is a better density model or that addressed writes improve C2.

10 Patch E3 v3 — feature-local multimodal normal memory

10.1 Why v3 changed the memory

V3 directly asks whether a collection of local normal appearance densities can avoid the global mean-absorption problem. From two normal support images, deterministic cosine farthest-first selects eight 1024-D support-derived anchor keys. Every patch is assigned to its closest fixed key in normalized cosine geometry. Each mode owns independent diagonal Gaussian or NIG sufficient statistics; a newly verified normal sample updates only the modes receiving its selected patches.

$$\text{mode}(p) = \operatorname{argmax}_{k=1..K} \cos(h_p, \text{key}_k);\ K = 1 \text{ (global control) or } 8 \text{ (local)}.$$

A normal image distributes at most one effective evidence unit among selected patches/modes. Full 784-patch scoring is performed and top-eight surprise is pooled after patch-wise predictive evaluation.

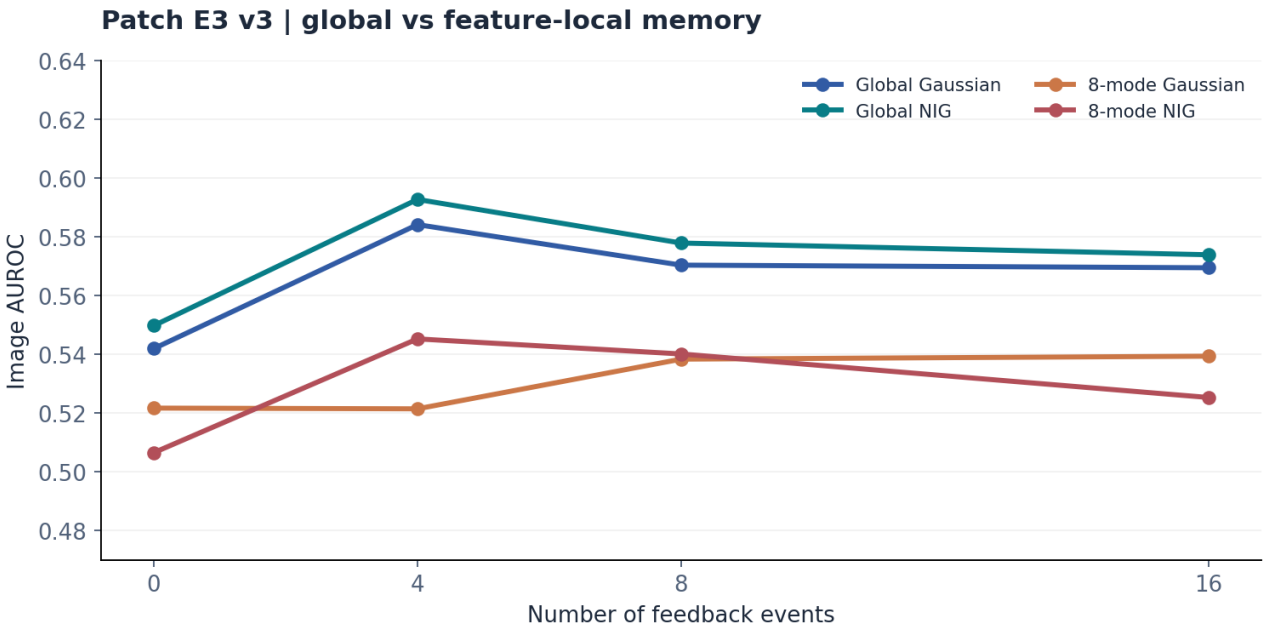
Query schedule: highest-surprise known-normal images according to a frozen local NIG selector, then the same top-32 selected patches for all four model branches. This choice is controlled, not deployable.

10.2 Four-way causal comparison

Branch	Memory structure	Density
Global Gaussian	One mode; all patches share parameters.	Diagonal plug-in Gaussian
Global NIG	One mode; all patches share parameters.	Diagonal Student-t predictive
Local Gaussian	Eight fixed cosine appearance modes; local writes.	Independent mode Gaussian
Local NIG	Eight fixed cosine appearance modes; local writes.	Independent mode NIG

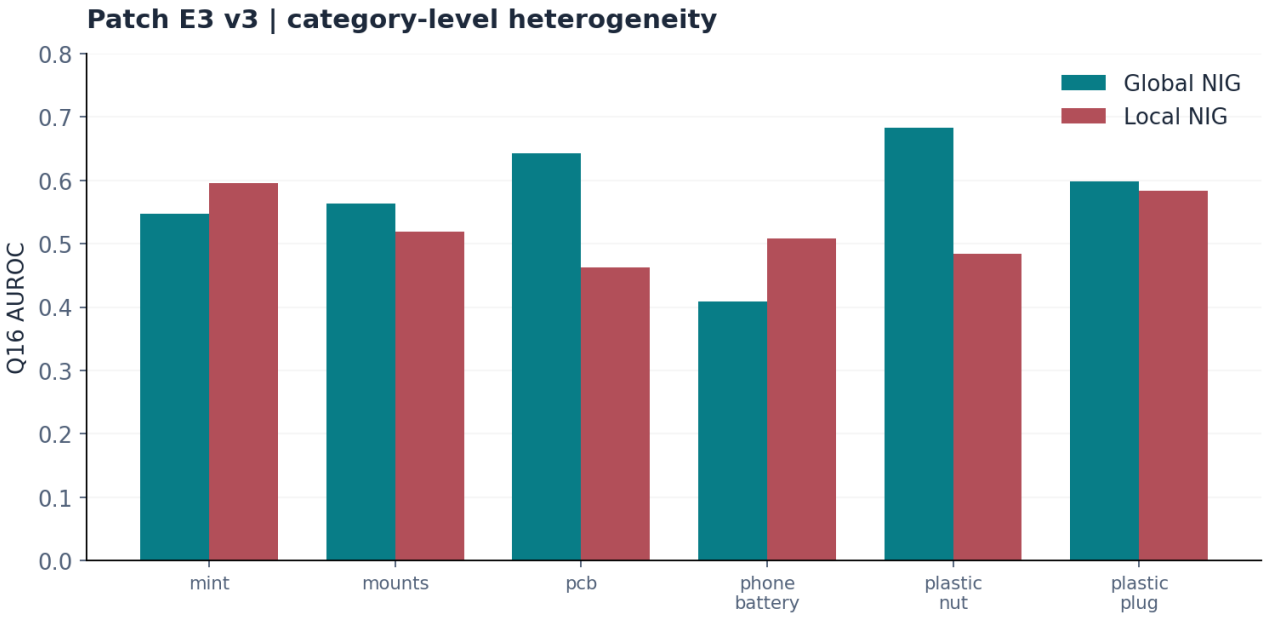
This design is a better test of feature-locality than v2, but the localization is hard cosine assignment into sparse support-derived clusters, not structured causal factorization. All eight local modes start from the same weak global prior and can receive drastically unequal evidence.

10.3 Results and heterogeneity



Mean AUROC on six source-CV products; global branches win at Q16. Local modes were not an automatic improvement.

Model	Q0	Q4	Q8	Q16	Δ 0→16
global_gaussian	0.5420	0.5841	0.5704	0.5695	+0.0275
global_nig	0.5498	0.5927	0.5779	0.5739	+0.0241
local_gaussian	0.5217	0.5215	0.5384	0.5394	+0.0177
local_nig	0.5066	0.5453	0.5401	0.5253	+0.0188



At Q16, local NIG is not consistently behind for every product, but the mean advantage is negative and the category responses vary strongly.

Local NIG category examples: mint improves 0.3861 → 0.5951, phone_battery deteriorates 0.6963 → 0.5088, and pcb deteriorates 0.5034 → 0.4619. Thus aggregate mean growth conceals opposite product-level responses. Only one source-CV fold/seed is available and no interval/calibration analysis analogous to original E3 has yet been reported.

10.4 Mode concentration and score response

Product	Largest mode share (Q16)	Normal score Δ	Defect score Δ
mint	56.4%	-0.1136	-0.1106
mounts	60.0%	-0.0620	-0.0606
pcb	72.5%	-0.1522	-0.1524
phone_battery	65.1%	-0.2340	-0.2491
plastic_nut	87.8%	-0.1102	-0.1091
plastic_plug	41.8%	-0.0839	-0.0843

For plastic_nut, one of eight modes carries ~87.8% of total effective evidence mass by Q16. This may reflect genuinely uneven patch prevalence, a poor initial partition, or high-surprise selection repeatedly choosing a single mode; the present JSON does not identify which. On average local NIG lowers both normal and defect image surprise to a similar extent (~−.126 and ~−.128). A score decrease is not automatically improved separation. In several categories, local NIG AUROC remains flat across multiple checkpoints while mode counts continue to change.

Current decision: Do not propagate this local NIG as the assumed canonical E4 memory. First diagnose assignment coverage, mode-wise posterior updates and paired changes in future patch surprise. The observed negative result is scientifically useful because it identifies a missing link between local state writes and future decisions.

11 Causal diagnosis: what failed, what remains untested

11.1 Four possible failure points

Failure hypothesis	Observational signature	Needed discriminator
Address mismatch	Updated modes rarely own high-surprise patches on future normal items.	Mode-by-mode feedback/sentinel coverage.
Predictive non-response	Correct mode receives evidence, NIG parameters move, but surprise barely changes.	Pre/post Student-t parameter and patch surprise trace.
Collateral absorption	Normal and defect patches assigned same mode receive comparable score reductions.	Matched change distribution for normal and defect-image patches; mask only offline for deeper diagnosis.
Top-k bottleneck	Many patch scores improve but the final top-eight set and image rank remain unchanged.	Pre/post top-eight identities, rank inversions, image AUROC.

11.2 Proposed E3-D diagnostic replay — NOT RUN

Replay the exact same E3 v3 feedback order, without changing model configuration. At Q0/Q4/Q8/Q16, store patch→mode assignments and mode occupancy, per-mode write mass and NIG posterior parameters (μ,κ,α,β), the predictive scale, patch-level pre/post surprise on never-updated sentinel normals/defects, and changes in the top-eight image-score contributors. Cross-tabulate whether a future patch assigned to a written mode actually benefits. Keep normal-image and defect-image patch groupings separate; without GT masks, “patch on a defective image” does not mean “defective patch”.

The intended E3-D result is mechanistic diagnosis, not a new SOTA metric. The experiment has been discussed but no result JSON or committed implementation has been supplied as of the evidence cut-off. If it reveals poor coverage, improve mode addressing; if coverage is good but surprise does not respond, investigate posterior scaling and the chosen likelihood; if both score classes shift together, investigate overly broad updates or insufficient defect localization.

12 Integrated evidence and the remaining research gates

Claim	Evidence now	What is still missing
C1 factor addressability	Cross-product raw DINO source information; E2 sidecars fail.	Repeatable learned/stable source-associated addresses; semantic factors not assumed.
C2 localized credit	Strong five-branch oracle-address gain.	Predicted routing and factor-specific normal memory updates outperform matched global controls.
C3 source-enriched feedback	E4B v1 tiny change; known-source upper bound higher.	Practical informative source feedback that improves future attribution and linked detector output.
C4 Bayesian normal governance	Image-level NIG calibration advantage.	Calibrated patch-level predictive model against stronger deterministic baseline and correlation controls.
C5 information-aware query	Not executed; E1B controller explicitly deferred.	Controller vs fixed/random query comparison at equal annotation cost.
Full system	No integrated positive end-to-end demonstration.	E4C with grounded spatial factor credit, five outer-fold target rotation, calibration and ablation.

12.1 Responsible next experiments, in order

First run E3-D replay to distinguish address coverage from posterior-predictive and top-k scoring failures. Second, revise patch-level E3 only according to that diagnosis; compare appropriate multi-modal and shrunk Gaussian alternatives with NIG using the same frozen features, supports and feedback events. Third, design a factor-/mode-local verified-normal E4A with explicit effective sample mass and collateral drift metrics. Fourth, connect defect-source feedback to a measured future output before making C3 detector-benefit claims. Only afterward open active query allocation (C5) and the five outer-fold confirmation.

12.2 What to say to a supervisor

“We have demonstrated that where sparse feedback is written matters in a controlled four-source routing experiment. Sparse masks also provide modest reusable spatial information. However, attempts to build compact address codes did not surpass raw DINO, and moving from pooled image-level NIG to local patch memories has revealed failure modes: local writes may not align with future high-surprise evidence and can lower normal and defect scores together. The current work is narrowing the precise conditions under which a frozen-backbone system can learn selectively, rather than claiming that the complete online architecture already works.”

13 Formula / metric glossary and reproducibility audit

Symbol / metric	Definition and how to interpret
h(t,p) / D	Frozen DINO feature of patch p of image t; D = 1024.
K / top-k	Number of selected suspicious patches for image-level score or feedback; k=8 for top surprise, 32 written in v3.
g / G	Address index / number of factor groups, G=4 Core-4 structural credit or 8 coordinate groups in patch v2.
r(t,g)	Responsibility weight that allocates feedback to address g; may be an uncalibrated softmax proxy.
α(t,g)	Actual evidence credit assigned to address g in an update; its sum controls total write mass.
n(g), sum(g)	Effective count and weighted descriptor sum for factor-specific mean memory.
μ, σ^2	Unknown normal mean and variance of one evidence coordinate.
μ_0, κ_0, α_0, β_0	NIG prior mean, prior strength, inverse-gamma shape and scale.
μ_n, κ_n, α_n, β_n	NIG posterior parameters after n effective observations.
ν, s^2	Student-t predictive degrees of freedom and predictive scale squared.
s(t,p), a(t,p)	Frozen detector anomaly score and Bayesian negative log predictive density (“surprise”).
Image AUROC	Probability that a random defect image ranks above a random normal image.
Macro-F1	Arithmetic average over AK/HS/QS/ZW per-source F1; not an image defect AUROC.
Normal FPR	Share of normal sentinel images scoring above a declared operating threshold.
Calibration error	|empirical 90% predictive coverage - 0.90| for held-out normal coordinates.
Collateral drift	Effect of a write on unrelated states or outputs; exact state locality does not ensure behavioral locality.

13.1 Branch & evidence files needed for replication

• [R1] Primary JSON: five_branch_structural_credit_audit.json (five branches; 40 source-side runs/representation).

• [R2] Primary JSON: e4_e4a_n0_audit.json; e4_e4b_n0_audit.json (4 source-CV groups × 3 support seeds).

• [R3] Primary JSON: patch_e3_audit.json (patch E3 v2; six products, CV0).

• [R4] Primary JSON: patch_e3_local_modes_audit.json (patch E3 v3; six products, CV0).

• [R5] Secondary handoff: Last_Paper_Session_Handoff_2026-10-07.docx (E1B, E2, original E3 metrics and decisions).

• [R6] Earlier handoff: last_paper_research_handoff_2026-09-15(1).docx (Stage-0 and Stage-1 controls).

• [R7] Repository experiment specs: docs/E2_ADDRESS_REPRESENTATION_EXPERIMENT.md; docs/E3_NORMAL_STATE_EXPERIMENT.md; docs/E4_ASYMMETRIC_FEEDBACK_EXPERIMENT.md.

• [R8] Current patch protocols: docs/PATCH_E3_E4_V2_PROTOCOL.md; docs/PATCH_E3_V3_LOCAL_MODES.md.

• [R9] Code for patch E3: scripts/abmg_patch_e3_e4_v2.py; scripts/abmg_patch_e3_local_modes_v3.py; scripts/abmg_e4_extract_evidence.py.

• [R10] Research formulation: attribution_aware_bayesian_memory_governance_step_by_step_guide(1).pdf and ABMG_extension_v2.pdf.

GitHub repository: https://github.com/jiangdizhao/ATTRIBUTION-AWARE-BAYESIAN-MEMORY-GOVERNANCE. Patch implementation branch inspected: stage2-patch-e3-e4-mechanism-validation, HEAD during preparation 0972ae0. Older branches and handoff commits refer to historical protocol snapshots; outputs do not always embed an exact producing Git commit.

Interpretation boundary: All percentages/averages in this document are conditional on their own protocol. Never compare E4 image AUROC directly with five-branch Core-4 F1 as if they measure the same object. New patch E3 results are not statistically validated across four source-CV folds or five outer folds.

14 Study checklist before resuming experiments

Use these questions to verify that the underlying research logic is clear before commissioning another implementation. A negative result should cause a specific hypothesis to be revised, not merely a broader hyperparameter search.

What makes the five-branch contrast causal? Branches share supports, stream, query indices and evaluation sentinels; write destination and total mass are explicitly controlled.

Why is the five-branch win an upper bound rather than a deployed system? The feedback includes a verified Core-4 address and an unusually useful correction opportunity, which a real active learner may not obtain.

Did E2 show that disentangled representations are impossible? No. It showed that seven tested 32-D sidecars failed against 1024-D raw DINO under the specified source-routing gate.

What exactly did original E3 NIG outperform? A simple image-pooled, diagonal plug-in Gaussian on sparse 90% interval coverage; the binary AUROC difference was tiny.

Why cannot E4B v1 establish detector benefit? The defect-source write changes routing sufficient statistics but is disconnected from the binary anomaly-score path.

How can eight local modes still behave like one? If high-surprise feedback repeatedly routes to a single key, effective write mass concentrates in that mode; most other modes receive no new evidence.

Why does lower surprise not guarantee progress? If normal and defect evidence are both absorbed, ranking/operating points may remain unchanged or degrade even as absolute likelihood increases.

What must E3-D observe to justify a code change? At minimum, patch-to-mode assignments, pre/post predictive scores, posterior parameter updates and the identity of patches controlling the image top-eight score.

What evidence would justify moving to E4? A patch-level normal-state mechanism that produces useful verified-normal corrections, preserved defect separation, bounded collateral changes and superiority to mass-matched alternative writes.

What remains the final generalization test? Freeze protocol choices, rotate all five outer folds so every Real-IAD product is evaluated once as a target; never choose hyperparameters using target outcomes.